Lab 03

DNN-AI331L

5G Quality of Service (QoS) Resource Allocation Prediction

Marks: 10

Objective

In this lab, you will build a complete Artificial Neural Network (ANN) pipeline using PyTorch to predict the percentage of network resources allocated to a 5G user. You will practice data exploration, preprocessing, categorical encoding, train/validation/test splitting, feature scaling, ANN construction, optimization, learning-rate scheduling, checkpointing, evaluation, visualization, and deployment through an interactive GUI.

Dataset

File: Quality of Service 5G.csv

Feature	Description
Application_Type	Type of application used by the 5G user
Signal_Strength	Strength of the 5G signal
Latency	Network latency
Required_Bandwidth	Bandwidth required by the application
Allocated_Bandwidth	Bandwidth allocated to the user
Resource_Allocation	Percentage of network resources allocated - Target

Prediction Target

Resource_Allocation (%):

Example hypothetical user: Application Type = Video Streaming, Signal Strength = −75 dBm, Latency = 15 ms, Required Bandwidth = 10 Mbps, Allocated Bandwidth = 8 Mbps. The ANN should produce a predicted resource allocation percentage.

Part A: Dataset and Preprocessing

1. Load Quality of Service 5G.csv using pandas.
2. Display the dataset shape, column names, data types, missing values, descriptive statistics, and sample records.
3. Select the five input features and identify Resource_Allocation as the target.
4. Handle missing or invalid values if present.
5. Encode Application_Type using one-hot encoding.
6. Split the data into 70% training, 15% validation, and 15% test sets.
7. Apply feature scaling. The scaler must be fitted only on the training data.
8. Convert the processed data into PyTorch tensors and DataLoaders.

Part B: Build the ANN

1. Build a fully connected ANN in PyTorch with an input layer, multiple hidden layers, ReLU activation, Batch Normalization, Dropout, and one output neuron.

2. Use appropriate loss functions
3. Use mini-batch training and GPU acceleration when CUDA is available.

Part C: Optimization Experiment

Train the same ANN architecture using the following three optimization configurations. Keep the dataset split and model architecture fixed so that the comparison is meaningful.

Experiment	Optimizer	Learning-Rate Scheduler
1	Adam	ReduceLROnPlateau
2	AdamW	CosineAnnealingLR
3	SGD + Momentum	StepLR

1. Train each configuration using mini-batches.
2. Use learning-rate scheduling.
3. Monitor validation loss during training.
4. Save the best model checkpoint for each configuration.
5. Implement early stopping to reduce unnecessary training.
6. Record training loss, validation loss, learning rate, best epoch, and training time.

Part D: Training Visualization

1. Plot training and validation loss curves for all optimization configurations.
2. Create a comparison table showing each optimizer, best validation MSE, best epoch, and training time.
3. Discuss whether the curves indicate underfitting or overfitting.

Part E: Final Evaluation

1. Select the final checkpoint using the predefined experimental criterion: lowest validation MSE.
2. Evaluate the selected model only once on the test set.
3. Report MAE, MSE, RMSE, and R^2.
4. Plot Actual vs Predicted Resource Allocation.
5. Create a residual/error visualization and briefly interpret the results.

Part F: Interactive Prediction GUI

1. Create a simple Tkinter GUI for 5G QoS prediction.
2. The GUI must accept Application Type, Signal Strength, Latency, Required Bandwidth, and Allocated Bandwidth.
3. Load the saved best ANN checkpoint when the application starts. Do not retrain the model.
4. Load and reuse the same feature encoding and scaling information used during training.
5. Validate user input and display a clear error message for invalid values.
6. Include Predict and Clear buttons.
7. Display the result in the format: Predicted Resource Allocation: XX.XX%.

Expected GUI Test Case

Input	Values
Application Type	Video Streaming
Signal Strength	-75 dBm
Latency	15 ms
Required Bandwidth	10 Mbps
Allocated Bandwidth	8 Mbps
Output	Predicted Resource Allocation: XX.XX%

Submission Requirements

1. Completed Jupyter Notebook (.ipynb)
2. Saved best ANN checkpoint (.pth)
3. One pdf file contains following:
    - Training/validation loss plots
    - Optimizer comparison table
    - Actual vs Predicted plot
    - Residual analysis plot
    - Screenshots of the working prediction GUI

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